

BI Dashboard Testing – Automation Strategy (Q3 2025 Plan)



Page is about?

Objective

To automate the validation of BI dashboards by comparing report data with the source system (database), ensuring accuracy, performance, and reliability while reducing manual verification effort.

Testing Categories to Automate

Type	Focus Areas
UI Testing	Validate layout, labels, colors, loading time, drop-downs, pagination, and export buttons
Functional Testing	Filters, prompts, drilldowns, sorting, date ranges, clickable actions, report logic per user/group
Data Validation	Row counts, aggregates (SUM, AVG), join validation, complex SQL result match, rule-based transformations
Security Testing	SSO login, user-specific data, embedded view access, report access restrictions
Performance	Stress/load testing under concurrent usage, dashboard load time, DB call latency

Recommended Solutions – End-to-End

Solution 1: Selenium + Java Hybrid Framework for UI/Functional Testing

Use Case:

- Validate visual layout, filters, prompts, and clickable components of BI dashboards (Power BI/Tableau/Looker-like tools).

Stack:

- Selenium + TestNG + Maven
- POM-based reusable design
- Allure/Extent for reports
- Grid setup for cross-browser
- Jenkins CI integration
- Supports parallel, batch, and scheduled runs

Features:

- Environment-based run support (DEV/STAGE/PROD)
- Screenshot + log on failure
- Reusable JSON/YAML/Excel for test data
- Generic keyword library for dashboards

Example:

```
Assert.assertEquals(getElementText("TotalAssetsCard"), dbResultTotalAssets);
```

Solution 2: ICEDQ – Enterprise-Grade Data Testing Tool

Use Case:

- Compare dashboard metrics to SQL queries in source warehouse.
- Detect missing or inconsistent records, NULLs, duplicates, mismatches.

Features:

- Zero-code UI
- Data comparison rules
- SQL query orchestration and validation
- Audit history + scheduling
- Workflow-based design

Pros:

- Quick to configure
- Great for non-tech QA & data engineers
- Email alerts for mismatch
- Supports JDBC/ODBC, APIs, SFTP, Snowflake, Oracle, etc.

Solution 3: DataGaps BI Validator**Use Case:**

- Automated UI + Data Testing for BI dashboards from Power BI, OBIEE, Tableau, MicroStrategy, etc.

Key Capabilities:

- Report-to-DB data validation
- Filter-level data matching
- Snapshot testing
- Automated regression tracking for BI releases
- Report rendering time

Pros:

- Specialized for BI validation
- Good documentation & scheduling
- Works with Azure, Redshift, Snowflake

Solution 4: Custom SQL-based Test Automation Framework**Use Case:**

- Lightweight DB validation framework using:
 - JDBC/Oracle drivers
 - Pre-written test SQLs
 - Assertion engine (Java/Python based)

Stack:

- Java + JDBC + TestNG
- YAML/Excel test suite
- Queries stored in version control
- CI integration with Jenkins
- Minimal UI required

Sample Test Case Flow:

1. Connect to DB.
2. Run expected and actual queries.
3. Compare values, print delta report.
4. Mark pass/fail.

Solution 5: JMeter (Performance for BI Dashboards)**Use Case:**

- Stress testing dashboards that consume API calls or DB hits.
- Simulate concurrent users, evaluate dashboard load time.

Approach:

- Use HTTP samplers to hit dashboard endpoints (if exposed).
- JDBC sampler for querying DB under concurrent threads.
- Include timers, listeners, and assertions.

Pros:

- Open-source
- Scalable via Taurus or BlazeMeter
- Report generation: HTML, JSON

Solution 6: Wiiisdom Ops (for Tableau/Power BI governance)

Enterprise Governance Tool:

- Automates functional + regression + performance testing
- Schedule BI regression validation per release
- Embed in CI/CD pipeline

Especially useful if:

- You're running **enterprise dashboards on Tableau, Power BI Cloud**
- Need validation of **permissions, visuals, bookmarks, metrics**

Recommended Plan – Execution Roadmap

Phase	Duration	Deliverables
Phase 1 – POC	1–2 Weeks	Pick 1–2 dashboards, write SQL validation, create 5 test cases with Selenium or ICEDQ
Phase 2 – Setup	2–3 Weeks	Framework baseline (UI + Data layer), Jenkins jobs, environment switch support
Phase 3 – Scale	4–6 Weeks	Add 30–40 test scenarios (high-priority), enable performance tests, alert hooks on failure
Phase 4 – Report	Ongoing	Weekly dashboard of BI test pass/fail, trend on mismatches, code coverage report, JIRA defects

Security, Permissions, and Access

- Create **BI Test User Profiles** for:
 - Admin view
 - Role-based access (Manager vs Analyst)
- Enable Single Sign-On (SSO) testing where applicable
- Validate:
 - Row-level security
 - Report restriction
 - Audit log consistency

Tools Comparison Summary

Tool/Framework	UI Test	Data Validation	CI/CD	Dashboard	Best For
Selenium + Java					UI/Functional
ICEDQ					Enterprise DB-to-report match
DataGaps					BI-heavy tools
Custom SQL Test					Lightweight DIY DB test
JMeter					Performance
Wiiisdom Ops					Tableau/Power BI cloud

Final Recommendation

- **Short Term (0–1 month):**
Implement a hybrid model: Selenium UI checks + Custom SQL automation for 2–3 high-use dashboards.
- **Mid-Term (1–3 months):**
Integrate **ICEDQ** or **DataGaps** for wider DB comparison and regression coverage. Add Jenkins email alerts + dashboards.
- **Long-Term (3+ months):**
Introduce performance testing via JMeter, CI blockers for data mismatch, Confluence dashboard reports, and full security testing automation.